

Yuchong Pan

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EDUCATION

- **Massachusetts Institute of Technology** Cambridge, MA
Ph.D. Candidate, Applied Mathematics (GPA: 5.0/5.0) September 2021 – May 2027 (expected)
 - **Research Interests:** Algorithms, combinatorics, optimization, graph theory, theoretical computer science
 - **Thesis Advisor:** Michel X. Goemans
- **University of British Columbia** Vancouver, BC
B.Sc., Honours Computer Science and Mathematics, with Distinction (GPA: 94.4%) September 2017 – May 2021
 - **Thesis:** *Optimization Problems on Network Flows with Degree Constraints*, advised by F. Bruce Shepherd

SELECTED PUBLICATIONS AND PREPRINTS

- A Better-Than-3 Approximation Algorithm for Demand Matching via Knapsack Intersection LP and Contention Resolution (with Michel X. Goemans). *arXiv preprint arXiv:2609.17932*. [arXiv]
- Bicriteria Approximation Algorithms for Demand Matching (with Michel X. Goemans). *arXiv preprint arXiv:2608.04223*. [arXiv]
- Planarity via Spanning Tree Number: A Linear-Algebraic Criterion (with Alan Bu). *SIAM Journal on Discrete Mathematics*, 39(2):728–751, 2025. [Journal]
- On High-Value and High-Flow Cycles at Basic Feasible Solutions of the Subtour Elimination Relaxations for the Symmetric and Asymmetric Traveling Salesman Problems (with Michel X. Goemans). *To submit*.
- A Counterexample to Box-Half-Integrality of the Intersection of Crossing Submodular Flow Systems (with Michel X. Goemans). *Unpublished*. [PDF]

SELECTED EXPERIENCES

- **Microsoft** Redmond, WA
Software Engineer Intern (Cloud and Artificial Intelligence) June 2019 – August 2019
 - **.NET Core Uninstall Tool:** Developed a guided tool that enables the controlled clean-up of a system such that only the desired versions of .NET Core SDKs and Runtimes remain. Conducted user research with project managers. Prepared user documentation. Released as an open source command-line tool by Microsoft to external users (770 stars and 72 forks on Github as of September 3, 2026).
- **Microsoft** Vancouver, BC
Software Engineer Intern (The Garage) May 2018 – August 2018
 - **Earth Lens:** Developed an iPad app with a team of 7 interns that identifies, tracks and analyzes objects in aerial imagery with local machine learning models to assist in disaster relief and environmental conservation. Released as an open source project by Microsoft to external users (52 stars and 19 forks on GitHub as of September 3, 2026).

SELECTED AWARDS AND FELLOWSHIPS

MIT Mathematics Department Fellowship (2021–2026). Work Learn International Undergraduate Award, UBC (2021). Trek Excellence Scholarship, UBC (2018, 2019, 2020). Silver Medal, National Olympiad in Informatics Top Competition, China (2015). Bronze Medal, Asia-Pacific Informatics Olympiad (2015). First Prize, National Olympiad in Informatics in Provinces, China (2013, 2014).

PROFESSIONAL SERVICES

Journal and conference referee: Operations Research Letters. Mathematical Programming, Series B. Conference on Integer Programming and Combinatorial Optimization. SIAM Journal on Discrete Mathematics. ACM-SIAM Symposium on Discrete Algorithms.

PROGRAMMING SKILLS

Python, Julia, Java, C#, SQL, MATLAB, Go, JavaScript, C++, L^AT_EX.